

ActInf MorphStream 002.1 ~ Alon Loeffler "From Neuromorphic Nanowire Networks to Neurons"

Source: [YouTube](#) Playlist: MorphStream Duration: 1:09:08 | Uploaded:
Unknown Transcript method: auto_caption

okay hello welcome everybody to the active inference institute's second episode of our MorphStream Series so these sessions are intended as a space to Showcase and discuss the intersection between neuromorphic Computing so brain inspired Computing and active inference um if you're new to neuromorphic Computing I did give a short primer in the first session which is available on YouTube so the format is that our guest speaker will present in the first part then there's a Q&A and then a wider conversation for the rest of the call so our first ones featured uh David Capel who's currently joined us again today thank you David and he gave an absolutely fascinating talk about um a free energy model to uncover the role of noise in the brain so it's like active influence at the synapse and today's episode features Alan Ller he's a postdoctoral scientist at cortical labs in Melbourne Australia cortical Labs being the company behind the very interesting dish brain paper which recently came out so he specializes in synthetic biological intelligence and has a PHD in neuromorphic Nano WI networks from the University of Sydney he also has almost three sorry over three years of experience in designing learning tasks and algorithms for brain inspired systems and today he's going to be presenting some really interesting research on neuromorphic nanowire networks and so if you want to share your screen absolutely thank you all right can you see this all good yeah all good all good fantastic so thank you very much for having me um and thanks for the wonderful introduction um not much more to say about myself about so that was perfect summary um yeah today I'm going to talk mostly about my PhD research which I finished last year from the University of Sydney which was in uh neuromorphic Nano networks and that's quite a mouthful we'll definitely get into what that is don't worry hopefully by the end you'll be expert um and I'll also touch a bit at the end about what I'm working on at cortical Labs at the moment and there's quite an interesting link they're slightly different but hopefully we'll get through that so just a bit of an O overview of how today is going to go I'm going to give a very brief background I imagine a lot of the people who are interested in this

kind of topics know quite a lot about AI and the state of uh AI at the moment but just in case you don't I'll give some background and and go over that um EXP what Nano networks are explain how what we actually do how we model these systems and you know how we actually apply these systems and then at the end just give a brief overview of what I've been working on for the last year or so so let's start with a bit of a background so the brain the the original inspiry or inspiration for artificial neural networks AI kind of this machine learning idea behind AI was the neuron um the most basic Ann is called a perceptron and it was sort of in the 50s and 60s it was um created to mimic the idea of this inputs into a neuron from this sort of dendrites into the through the cell body into the axon and then across as an output into different neurons and that's sort of the idea that we have that's sort of what became this beginning of uh artificial neural and AI in general um and so but but there's there's quite a few key differences between neurons and the structure of neurons in the brain and how we implement our perceptron Ann interpretation of these neurons mainly there's the idea behind vum AR vman architecture which is what how we pretty much implement all of our Computing these days compared to how the is structured uh so in most modern AI systems we implement this uh software of AI on a hardware Von Norman Hardware where we have this separation between Computing processing and memory and we have to bus information back and forth from the CPU to the memory which is quite a big bottleneck it takes quite a lot of time to do that and that's pretty much the main kind of limitation of of this uh architecture whereas in the brain there's sort of this integrated idea of memory and processing that happens at in the synaptic kind of uh gaps between neurons in in neural cells in chemical processes all that kind of stuff occurs simultaneously and so we're trying to overcome this bottleneck of this busing information back and forth which actually takes quite a lot of energy and quite a lot of time the main way we've been doing this is adding more transistors this is sort of how the structures are built it using transistors more of those transistors are added so that we can have more and more computer processing this is what's called Moors law and you may have heard of that before but the idea is over the past few decades from the beginning of the perceptron which I introduced we've sort of seen this two-year doubling idea of computer processing power and energy requirement and then in the last sort of decade or so we've seen that this is not a linear well still linear but it's not quite as simple this relationship it's not a doubling of energy and power we're seeing this 3.5 3.4 month doubling of requirement for power um so more and more and more transistors are required to do AI and Ann kind of uh processing and more and more power is therefore required which is a big issue why is this a big issue uh well

firstly I'll get i'll get back to that in in a bit of a second but importantly um another important aspect between you know the brain and well Norman architecture another difference is this brain circuitry idea um and so while modern applications of AI are really really really good at uh you know um uh object recognition and patent recognition uh they've able to beat humans in lots of things like that but their original uron inspired architecture which I discussed from the perceptron has transformed into much more complicated system that's reliant on on these higher and higher power requirements and energy requirements and are held back by this bottleneck um and the feed forward structure that we see that we all quite familiar with where you have this input layers into hidden layers uh into an output layer where information Flows In from the left goes through these layers and out to the right it's an inherent approach in AI there is some recurrence that can be introduced but even that's quite a simple um method and this is highly highly different to biological neural networks uh we see a very different structure we see a very different kind of functionality even in these recurrent networks and so the question is you know humans behave at least sometimes intelligently with significantly low energy requirements you know you and I only need few meals a day worth of energy the brain works at around 20 watts of information of energy sorry um and in order to try and capture this kind of intelligence and and processing capacity of the brain um we need to try and overcome these bottl necks and structural issues that I've mentioned briefly that AI faces and so to do that we can turn to the brain for inspiration um and then the question that I was asking during my PhD is how can we capture physical nature of biological intelligence so that's a brief background of where the state we're in and this is where neuromorphics enters right that's the background behind what we were what the state we're in but neuromorphics is one suggestion into how to um solve this so neuromorphics the term neuro comes from neuron morphic meaning mimicking or copying uh neuromorphics uh sorted into two main uh categories one is the idea of neuromorphic chips which is something that you is probably more familiar to uh similar to you know uh computer chips uh seos electronic electrical chips stuff like that and they're used as mainly AI accelerators so how do we Implement current AI models the software on this new hardware um to get more neural mimicking behavior and redu power requirements and something that other people might be more familiar with is spiking neural networks where the idea is we take traditional artificial neuron networks and we turn them into this binary input output or binary inputs and binary outputs where it's just ones and zeros and so you get significantly reduced amount of data that you have to deal with the weights are significantly um more simple to deal with and so this is a much easier kind of uh

data set to deal with and uses a lot more a lot less energy as well on the other hand we have this memristive systems and so memristive systems are systems that have this thing called conductance-based synapses uh where the the idea behind it the word Memristive comes from Mistor which is a variable resistor and pretty much it exhibits memory as the kind of current a hysteresis in a current response curve that might not mean anything but I'll explain that to you in a little bit pretty much the idea is that we get these memristive systems that are much more uh biological like when they're self assembled much more array likee when they're array assembled but they have this more uh nonvolatile they have this more interesting Behavior than neuromorphic chips do so there's some examples of this these type of systems uh Intel and IBM mainly work on AI accelerators in ly here and true north um through the human brain um through human brain project we have developed uh Spa brain scales and a few other models to try and Implement spiking neuron networks which try and represent replicate spikes in the brain neurons and then memories of crossbar arrays this idea where you have these sort of metal metal bottom and top electrodes you have this cell in the middle and you get flow of energy electricity through that I'll explain a lot more about this later it's just sort of a bit of an the thing that we are focusing on today though is something called Nano networks which fall into the self assembled networks which means they assemble into a network from the bottom up rather than what you might imagine these other three categories do in this kind of top down clean lab fabrication um e experience and that makes them a lot more similar to biological systems so that's a a big mouthful but we'll get we'll break it down a bit more so what are nano networks well Nano networks are typically made out of a a simple material like a silver material um they're Nano so they're really really tiny on the orders of nanometers in width about a thousandth of a hair width of a hair um and they get to about five to 10 micrometers in length so they're really long and very very thin quite similar to to a neuron a neuron and on the right here you see an example of what a neuro a nano Network looks like a silver Nano Network so made out of silver material compared to an example of what a biological system look like so there there's quite similarity quite significant similarities in the structure of how they look um so what happens is we deposit these Nano these these silver wires onto a glass substrate usually and they self assemble so they assemble into this network and they start to cross over each other so you can see wherever um let me Point get the pointer out where is the pointer sorry there we go so wherever a wire crosses over another wire we assume that there is a junction or aapse similar to you know where a dendrite axon connects into a dendrite in neural structures and so the idea behind this the structure is

resembles itself and the function also resembles itself that's what these two graphs at the bottom look like so when we zoom into these kind of uh Junctions as you can see here that's a zoom in of that we get this uh silver two silver wires are represented here we get this in um insulating polymer in between it's just a simple plastic called PVP polymer um in between and then when that is off meaning when there's no electric current being passed into the system nothing special happens the W the wires aren't actually touching each other there's no um electrical conductivity but then when we turn it on when we feed in some kind of input voltage input pulse like is shown here we see this ion migration this flow of atoms positively charged atoms from one of the wires to the other through this plastic polymer and this is what we call like a switch it's it's similar to what an action potential kind of looks like where action potentials occur and then you get this uh increase in conductivity in the membrane of the neuron we see a similar behavior in nanowires this is sort of the basic blocks of how neurons communicate with each other for those of you who don't know how that uh happens um and we see a very very similar kind of response occurring in these uh Nano now I mentioned memory before mistas have this property of when this uh switch like Behavior occurs when you get this uh nanofilament forming between across here uh you get it's a nonlinear Activation so it's not predictable it's quite a difficult thing to model and when it forms it has memory for previous times previous signals that have occurred that have been input into the system so it forms at a certain Pace nonlinear Pace once we stop that pulse it starts to Decay slowly back across the network at a different rate than what it originally uh formed so you get this these multiple complex behaviors occurring and because of that decay you slowly like uh the the ions Decay very slowly and imagine you've stopped the signal for a couple of minutes the the bridges halfway and then you you start the signal again that bridge is going to form a lot quicker than a bridge that hasn't been formed previously so that's where we get this memory aspect from it that's a junction level memory we also get all the beautiful recurrent um behaviors of you know information isn't just free forward um forward blowing like in um like in artificial neural networks we get this kind of uh recurrent activity that information flows in all directions in these networks which is really really important for information processing biological like information processing okay I've talked a lot more than I thought I would so um let's power through um you can take your time it's no problem great one really important really interesting thing about Nano wise is we can model them in uh graphical representation where we take the wires and we represent them as nodes the these actual wires and whenever there's a sinapse or a crossover between the Junctions we can res represent those as edges and so that really helps

us to look at the structural properties and also the functional connectivity of these networks so this is an example of what we've done with a model you can take a real network and you can model the density you can model the length of the wires which we can measure relatively difficultly but we can measure them and then you can construct a simulated version of it which is definitely not one to one that's one limitation that we have is very difficult to model exactly the same but it is experimentally validated meaning that the activity that we see in the simulated network we see similar conductance similar current similar voltage in the real nano and what's really lovely and what I've published in first PhD paper is you can model these uh these nano in a graphical representation as I mentioned before and then compare them to other structures that occur in the real world for example a fully random network over here a completely lattice like network where each node is just connected to its nearest neighbor and then our NY network compared to a very simple biological system which is called C elegans so what we see is that our biological nano network is much more similar sorry our bioinspired neuromorphic nano network is much more similar to biological systems like the C elegans than it is to random networks or these lattice like objects and these lattice sort of like structures is what we see in typically in CMOS chips or computer chips um uh a bit more about the modeling I've explained a bit about how this conductivity uh this low to high conductance occurs but we can model that using a mathematical equation for those of you who like maths this is the only equation that I will share in this presentation I don't like having uh equations on presentations but it's a pretty cool thing to see pretty much we can model when the voltage is higher than what we have V_{set} state which is shown in this graph here so the voltage is on the x-axis the current and the y-axis once there's a voltage set once we reach a V_{set} and once our voltage is higher than that we have this kind of switch that is on so we get in a high conductive phase and then when the derivative of this Lambda function which models this kind of filament is at zero we have and when the voltage is between uh this set state and the reset state that we have we get this kind of activity where it slowly decreases the current decreases and flips and then when the voltage is lower than reset we get back to the off state where the is at zero so this is what I was mentioning mistas have this pinched hysteresis curve that's what you see here this is called the pinched hysteresis curve and this is sort of the idea with where we want to see this kind of uh nonlinear memristive activity that's what makes a nanowire a neuromorphic system interesting um don't worry about it if that's too much jargon for you it's just a bonus for maths whoever likes the maths in here um so a bit more of a visual explanation of how

this what the nanowire network is actually doing this is a model of the the network we've simulated um where we have this input electrode up here which is represented by this green uh star and then we have an a drain or an output electrode bottom right here we feeding in a voltage uh a standard voltage bias simple DC voltage bias and we start to see each one of these wires as I mention each one of the wires is me is modeled by a DOT here and each one of the connections between the wires is modeled by an edge and you start to see this kind of increase in current flow that's what this is showing current flow across the network from uh the the source to the drain you see this what what we call a winner takes all pathway where the a path is formed where these Junctions switch on so all of these Junctions are at that on high resistance State and all the others are not yet and then you start to see like more parallel paths beginning to activate once that original path is um formed which is a really interesting um functionality that we can model we also see this network level kind of increase in G G is conductance so it kind of models the it's the inverse of the resistance so pretty much we start at a very very high resistance State and then we or low conductance and then at some point and that point is at the the path of lease resistance when the drain and the the source and the drain are connected we see this massive increase in conductivity massive decrease in resistance and that's that swap between low resistance to a high resistance State we see that at a network level as well as an individual Junction level which is really interesting behavior um this is this sort of nonlinear activation function which might look familiar to some of you that have played with artificial intelligence before for those of you who haven't this nonlinear nonlinearity is sort of how we do our um calculations and transformations in um hidden States it's you know it's this there's different types of nonlinear activation functions but there's the retic and linear activation functions anyway not important this is sort of what this system does inherently we don't have to train it to do that it just that is the inherent properties of this network um whereas in uh artificial new networks we sort of have to play around to get what the best model is to fit that um the memory aspect which I mentioned also happens at a network level not just a junction level so if you remember I mentioned that each filament has memory for what happened previously the network also has that memory and what you can see is when we're feeding uh a bunch of uh short small pulses for a short amount of time you get this increasing conductance at the network level PR what I explained previously and then it sort of resets back to its original state after a certain amount of time uh that's because each one of those Junctions that I've mentioned here that is activated takes a while to Decay and go back to its normal State and eventually it all goes back to normal but if you increase that uh the number the

amount of time that you're in inputting this voltage and the number of voltage pulses you can get the network into this kind of steady state where it doesn't reset for quite a long time and so you see this full Network kind of activation level similar to longterm memory or long-term potentiation compared to short-term memory these are just some really cool um things that we've seen in the network it's you know much more complex than than I have time to really delve into it but I just if you have more questions please feel free to ask I can I can answer as many as as needed um and so a lot of background a lot of kind of information dump but what can we actually do with these what is why are these cool you know why are we even interested in these well firstly we have to understand how we apply these so as I said we take these Silver nanowires And We dropcast Them meaning we take a little pet we take um from a pre um created mixture of these PVP um silver Nano networks and we put them in the middle of an electrode array that we've set up so an electrode AR a might look like this where we have nine electrodes on one side that are measuring the activity from electrical activity from the networks and nine on the other side and you can kind of program any of these to do anything that you want nine and N is quite small compared to a lot of um electrodes but this is just a start that what we did in our system um and so one one potential approach to using these n networks is something called Reservoir Computing that's a bit closer to uh artificial neuronet Network kind of implementations uh the idea is you have your inputs like an artificial neural network you have your inputs you have your hidden layers in the middle and then you have your outputs the main difference is in Reservoir Computing is the reservoir system or the in the in between layers are untouched we do not touch them in traditional AI the idea is you use something called back propagation to change your inputs to get your outputs you change the the the weights of your each one of these kind of nodes to try and match your outputs with a desired output that you have and you kind of if the desired output is not similar to your current output you back propagate you change the weights of the middle layers and then you try and make that a bit closer to the output that's how traditional amn's work in Reservoir Computing we just let the network kind of do its thing we don't change the weights we just let the inherent physics um do what it wants and then we train an out on the outputs we train a very simple model to try and understand the outputs from the network and so we can do the same with physical systems like n Nano networks where we just take out a traditional Reservoir system and we put in our Nano networks so what does that look like well in a task which I did in um well a bunch of my collaborators did prior to me and also I did in my PhD we have this input signal it's called the nonlinear transformation task we feed in an a sinusoidal inpo input just a

sign a simple sine wave to the reservoir system which is our Nano networks we let them do their thing we don't touch them they have some Physics that they do based on all that kind of interesting stuff that I explained before we get the linear readouts from that Network and we uh train some weight Matrix to try and um using linear regression very very simple kind of method to try and convert this input signal into a different output signal and the only way that that can happen is if the reservoir system in the middle has nonlinear Behavior interesting kind of um behaviors pretty much um to try and that that that can transform that input signal inherently and that we don't need to train a very complex system to do um the other task for example is something called a memory capacity task where you have a random input signal that you're feeding into the reservoir um at each time point and then you you delay your signal by however many steps and then you're trying and Rec trying to reconstruct the time Point end steps backwards these are just a couple of examples don't worry if it's a bit too confusing but um so what did I do another thing that I did which was really interesting in my second paper was trying to combine these kind of interesting structures of Nano networks which make them a lot more biological with with this res our Computing uh model and so over here we have a nanowire network all of these use the same Nano Network model but we just changed the the structure so this is an example of a fully connected uh two layer Nano network not nano Network sorry artificial neural network uh fully random Network a nano Network and a biological Network called the celegans and we did we performed these two tasks and we we showed that the Nano networks at certain values outperform if not perform as well as other um types of networks without the need for these and most interestingly is these bipartite networks which we use or these Ann type structures perform the least effectively uh when using our models so it shows that there is an importance in this biological structure as much as there is in the biological function which I've been talking about previously so this this kind of combination of the two is the most important animportant aspect and uh lastly last thing I want to talk about about the Nano networks this is a very speedrun introduction into it was my final paper in my PhD which was actually trying to take brain inspired um learning tasks and Implement them in this biological intelligence so Reservoir Computing was one way to try and Implement some kind of AI like task into this but that's not the point of this talk and the point of my project was to try and get a lot more brain inspired learning a lot more brain inspired intelligence those kind of aspects and so that brings us to this type of task where I try to implement a brain inspired learning task uh inspired by a psychological task called the endb back task where you give a Target pattern to uh participants then you give them a bunch of

different patterns and then you ask them after a certain number of steps do you remember the original pattern that I presented to you and that's what we did to this network we presented a specific pattern for example specific pattern number a pattern a and then we gave it a bunch of different patterns and then we asked it we wanted to see how many steps back it can remember that original step that original pattern that's pretty much and if it if it uh recognized the pattern successfully we uh gave it some external feedback so we either punished it or rewarded it and um if it that was in the this what we call the physical reinforcement um category and if we had no reinforcement then we didn't punish it at all we didn't give it any feedback and we just saw what it would do and so the results so this is an example of what it looks like a bit more um clearly so you if our Target pattern here is pattern a we give a bunch of patterns then we give pattern a then we give a couple of other patterns and then we ask for pattern a again so this would be three steps back where n equals three so the N back would be three and this would be kind of what we feed into the network on the left is where the inputs are on the right is sort of what the output that we wanted to get associated with and each one of these patterns has a different output that is associated with it and what we saw is that the accuracy this is a bit of a comp a bit of a you know difficult thing to look at but pretty much the square the square um dots are what we saw experimentally the circle dots uh what we saw in simulation and the reinforcement so the the what we see with with the reinforcement so giving them reward and punishment is at the top here and what we see without giving them a reward and punishment is at the bottom so interestingly enough at this is the number of steps back where we presented that original um pattern so for example at n back equals seven the original pattern would be right at the start and there' be a bunch of other patterns and then we'd ask it again and so what we see is that without any reinforcement there is this gradual decay of memory over time for that pattern and that makes a lot of sense humans are pretty bad at this task as well we forget quite easily after about seven steps as well so that's quite an interesting thing but what's really interesting is when we implement this reward and Punishment uh application which is based in biological reward and Punishment something called supervised learning um and reinforcement learning these two things combined we see a massive increase in with in memory for this uh patternn and this this um this uh yeah the target um importantly we also saw that activity when we had two Targets so if we had like a target a and then we swapped and asked for Target B it managed to remember both of those at the same time um which is really interesting we haven't pushed that to the limits but that's a really interesting aspect and so that shows that we can get this kind of biologically inspired learning mechanisms to give us some interesting activities in

this quite complex biological inspired system uh how does that look I've got a video hopefully that will work on me no oh yep yep yep we got there come on you can do it okay maybe I need to select [Music] the sorry pointer options turn off laser pointer play so this is an example of without the reinforcement we have our inputs on the right here which is a bit confusing and then we have our Target pattern this circle Target pattern so what you wanted what you'd want to see is a clear path from anywhere around here to this target meaning that the N the network is learning to to prioritize that pathway over time uh without reinforcement we're seeing that yes sometimes that path is being formed most likely when uh you know that signal is being implemented but a lot of the time it's it's not it's not really prioritizing and for on the other hand in the reinforcement Paradigm you'll see pretty quickly we start off with this sort of random kind of path activity and then you'll see that over time this path is remembered by the network and prioritized you do still get other Pathways but they're slowly kind of weakened over time but this path is almost always staying there and that's the idea behind this reinforcement learning and and so the network has remembered that whenever we're given the input signal we want to go to that output TP which is really cool to see I think this is a really cool video and there's a summary image of what happens over time this is flipped so the inputs in out on the left sorry I know it's annoying but um the pretty much the red regions are the regions that are strengthened between this when we give reinforcement and we don't give reinforcement and the blue regions are what are weakened so the network is effectively learning to to weaken these Pathways and strengthen these Pathways similar to what you might you know you might say it's similar to this pruning activity or strengthening activity in a synaptic kind of uh brain like activity and and strengthening of synapses which is really really interesting um really cool that we managed to see this so that's quite a lot of a lot of information how does this even link into what I'm doing now so that that was my PhD right now I'm working on real neurons at cortical labs and you know neuromorphics are really cool especially neuromorphic nany networks they try and replicate the brain they try and replicate well in a a much smaller scale neurons and synapses they do it in this nonlinear type of activation this really interesting structure and function um combination that mimics that of neurons and synapses but they're not neurons and synapses and so we know that that neurons and synapses and other types of cells lead to really cool stuff you know I'm sitting here and talking to you because my neurons and My Gal cells and my interneurons and all that kind of cool stuff works together to allow me to speak to think to breathe to do all the cool things that I can and we're not at the stage where neuromorphic Nano or other neuromorphic systems can

do that they can do a lot of really interesting stuff they can definitely reduce the amount of energy requirements that we have but we're definitely not at a stage where we can implement that so why not try do it in neurons we know that neurons can do it let's try and do that and that's what cortical Labs is trying to do so pretty much what we do very very briefly is we take um uh induced pluripotent stem cells we take a blood sample from a donor uh a human donor very very non-invasive simple procedure we take some blood like you would at a blood test or you know if you're donating blood and the amazing biologists in our team managed to convert it into something called induced pluripotent stem cells which means cells that can be reprogrammed to be any type of cell and we do this uh we change these ipscs in induced pluripotent stem cells into something called neuronal cultures uh or neural cells that are very much cells that are in the cortex very similar to cells that you and me have in fact if the blood sample is from you those cells would be exact would have the exact genome that you have so we've seen that in patients who have epilepsy if we take their blood the cells that we take also have epilepsy which is really interesting we can also take it from mice embryos or other types of animals which a lot of other people do and then what really is cool about and differentiates cortical Labs is we teach them to play games and we do that using the free energy principle which you know where I had to bring in some active inference related stuff and so that's sort of what we did in our first paper we presented uh we modeled this kind of pong environment the game pong hopefully most of you are um know this game and the neurons we tell the neurons we put them on this chip similar to what I showed in the nanos um and we tell them you are the paddle control the paddle based on this kind of structure this setup that we give you and try and predict where the ball's going to go and the way we do that is we give them either an unpredictable feedback or a predictable feedback um and when they miss the ball we give them this unpredictable random kind of feedback and when they hit the ball they get a predictable sensory feedback and this fits quite nicely with the free energy principle the idea that we're trying to minimize the randomness and try and get some predictable interesting behavior in in order to sort of maximize our reward and maximize our understanding of the environment uh we also give a silent feedback where nothing happens so the ball just continues just as a control um and what's really interesting and I didn't uh didn't actually include the graph but I can refer you all to the papers and I'm sure Brett the CEO of or the CSO of my company has spoken a couple of times here um so you may be familiar with that but we see that when we give this predictive predictable sensory feedback the neurons learn to play this game very very quickly within 5 to 20 minutes of playing the game and they play it

at statistically significant statistically better than a pardon me than average or than chance uh would expect and so this kind of encoding scheme where we're just giving this predictable versus unpredictable sensory feedback that's the only encoding that we're giving them we see quite interesting um you know support for the free energy principle being used in this kind of neural based system so there's a lot of stuff that I've cover today um there's still so many questions left I'm more than happy to answer those questions ask raise more questions um if anyone wants to ask me in person this is my email here ad me on LinkedIn I'm love to chat to anyone and everyone about this you know 40 minutes it took in the end um is not enough to cover everything that I did but I hope it gave a nice overview and a nice start for people to um go in and and try and understand and ask questions so thank you very much to my supervisor for my PhD my current bosses and and workmates and for you guys for having me and for listening and I yeah look forward to your questions thank you so much that was absolutely fascinating that was amazing um yeah I have so many different question I don't even know where to start I mean the the first the main thing I was wondering about is when you were showing the the networks with the mback test which was super interesting Nano networks and you were talking about reward and Punishment I was quite interesting how Implement that but then then you show the um the work at Cal Labs with pong and it's about unpredictable so is it a similar type of reward and punishment that you did with your simulation with your um Nan wire networks to that so unfortunately I hadn't I wasn't aware of uh active inference for energy principle all that cool stuff when I was doing my PhD that was I was introduced to that at cical Labs I would have loved to try that on these Nano I imagine it would be a fantastic way of doing this um we did something more similar to what you would imagine in gradient descent but also this kind of feedback so in a sense we're trying to give a more uh not maybe not predictable but we're trying to match what the output of the network is with what the expected output is so maybe that is in a little bit like a predictable kind of trying to increase the predictability pretty much what we did is we changed the can you see my mouse or do I need the um oh you can't see my okay great which pretty much what we did is we changed the voltage of the um the desired output electrode um and so if the correct paths were being uh formed because this is a physical system and it's a metal system at the end of the day you get this kind of path of lease resistance uh forming from where your sources where your inputs are to where your drain is the drain is pretty much just a grounding you know a zero voltage kind of thing and but you can play around with that voltage it doesn't have to be zero you can increase it you can decrease it you can increase and decrease the other ones and so we

implemented something similar to gradient descent on these uh voltages on these electrodes and when voltage is increased for example then you imagine there's less current or less current will flow to that um to that electrode and when voltage is decreased more current will flow to the electric so on a very basic level that's sort of what we implemented but we saw that wasn't enough that really wasn't enough to train it it needed this rewarding kind of thing so we needed to give some kind of threshold that when that's reached we increase it it was a bit more complex than that but that's pretty much the idea behind what we did thank you okay very interesting cool yeah does anyone else have any questions yeah I would have one question really thank you thank you um so you started your talk with um motivating it with the energy consumption of the brain correct really low so can you give us a kind of rough estimate of how much these nanobiosystems right they're tiny tiny tiny on the on the scales of Watts you know we're talking even even significantly lower than that um the only energy that's kind of being used in the system is what you pump in from the you know the the inputs that you give it so you have to give some kind of voltage as an input or you know and then some current is formed from that but the system itself the beauty of this is because of it's a you know it's a fully connected kind of physical system we can solve the weights automatically using kohn's laws of current and and and voltage it's it's just a physics problem at the end of the day and so physics just does it for us this the spreading out of the voltage the spreading out of the conduct of of all that stuff as I showed if I go back here right all this stuff that we see here the spreading out of this kind of activation happens inherently based on the structure of the network and that's why it's so important to try and understand that structure and combine it with a function so when you think about it that way that's significantly lower energy requirements than trying to change each weight individually of each one of these you know Junctions over and over and over and over again to better you know get a better result okay uh how many signups are typically could you realize in such a physical system for sure in our simulations we use significantly smaller simulations because that's what requires a lot of energy to simulate um we use about 100 to 700 wires in the actual system tens of thousands if not hundreds of thousands of of wires per per system the really other interesting thing is each one of these systems will be unique so you'll have a slightly different structure a slightly different kind of way that it's uh implemented and a lot of people think oh that would make it function significantly differently but we don't see that we see you know standard kind of characteristics obviously there's variance but you you know the same is with humans and with brain you see very different structures and very different you know experiences that humans

have and there is a variance in in our who we are but for much of the case our activity and our the way we function is quite similar at the end of the day okay yeah but um in in the systems you actually really realized physically um how many so I guess if you have like 10 thousands of wires um only a a subset of those would actually form correct functional potential synoptic connections right so there's probably are to are like electrically isolated from each other right so I guess that 100% so the really nice thing about these for now is that they're two-dimensional and so because they're two-dimensional a lot more of the wires cross over each other in kind of separated regions so you wouldn't you wouldn't get for example the right region connected to the left region right um and so that makes it a lot simpler to model when you start getting three-dimensional models you get this kind of you know wrapping around and a lot of different connections connecting to each other so there could be a lot more connections or a lot less connections it doesn't necessarily mean that when a wire crosses over another wire a perfect connection is formed so it's quite difficult to expect this and model it that's why we have to rely on uh electrical characterization and seeing these kind of conductance Curves here right comparing the conduct curves versus with the models that we have with the actual uh the actual physical systems that's how we do that kind of model okay yeah thank you and for the for the dish brain for the like biological cells so I mean the the brain is actually fairly structured right in the end yes correct so are you working in the direction of also using a bit more of the brains morphology Absolut cell types and also layers so we have a whole bunch of different cell types that we have we do all sorts of combinations of excitatory inhibitory cells gal cells asites all that kind of stuff um whereas currently still also working on two-dimensional layouts or two and a half dimensional layouts but there's we've had some collaborations with people for example John Hopkins University in America um which I believe I can talk about um hopefully I can talk about uh where we work with organoids um which are three-dimensional tiny little kind of cell structures they're not fully you know structured like what you'd see in any kind of brain organ um structure but they're a lot more similar to a tiny kind of brain structure and so what would really I mean my end goal in this is to try and stack chips together you know so a chip might be have a two-dimensional structure but then you can have it um touching another chip forming you know forming long connections across that kind of barrier with with another chip and then your first chip might act as some kind of sensory region and your second chip might be a processing region right and then you can form these Pudo structures in that way um what's beautiful about neurons compared to nanoes which I think is another reason and I didn't touch upon this is neurons self

organized after they're um placed down so with Nano eyes you put them down they self organize and they never move again you know they will never move again because they're silver wires and you cannot move neurons automatically are constantly looking for connections constantly changing their structure based on the input that is given to them so you know they will form inherently structures as well um and we're really we're exploring that we're exploring how that happens when you're giving them predictable feedback how that happens when you're giving them unpredictable feedback how it happens when you give them no feedback at all and these are all the kind of questions that we're trying to you know um answer at cortical Labs which is I think I find extremely interesting okay thank you it's interesting um I'm really interested in the applications which you also briefly touched on um in terms of when you said you get a donor with epilepsy and then you um their cells also exhibit epilepsy which then leads me on to thinking about well the applications of this could you potentially see this as sort of like a neural aesthetic or something where um say with someone with epilepsy they have that little part of their brain with their High concentration of epileptic cells removed sometimes I think that already happens now with EP epilepsy surgery could you see this almost of of them being an implant where you've taken their cells you've edited out the epileptic Gene if you if it was able to be identified then you reprogram them into neurons basically without epilepsy and then sort of put them back into the bit that you could take out I think that would be absolutely fascinating um I think there's so many different ways this can go down luckily I'm not the one who has to decide that I have a lot of ideas myself uh but uh I think one really key thing that we I should highlight is you know what what you mentioned is really really interesting and some people might think oh that's currently already being done in brain computer interfaces right there's a key difference between what we're doing and what brain computer interfacing is pretty much the idea with brain computer interfacing is you have a chip that reads the activity of the brain right and it spits out whatever that activity is and then you have some very clever algorithms to understand what that original signal was in the brain so for example you might think of a tennis game and then you get some kind of uh electrical signal that represents a tennis game and eventually these uh algorithms will pick up that you're thinking of a tennis game now that's really cool really important going to help a lot of people but there's a significant limitation in that whereas we cannot we we don't know what the brain is doing still to create those images right of a tennis game why are certain regions lighting up to represent that and and brain computer interfacing is not giving us any information about that you know it's just trying to reconstruct what is already being taking

place and what we're trying to do is we're actually trying to figure out the algorithms and the encoding schemes you know whether it's free energy principle whatever else might be hopefully it is something as straightforward as the free energy principle um where we want to actually write those encoding processes into these neurons teach them the output that we want them to give us rather than just interpret what the output is so we're doing the encoding into the networks and the decoding whereas the brain computer interface is sort of just the decoding and so you know in an epilepsy kind of um situation you might be able to have a chip like that in in the brain this is just you know pure conjecture but it could be interesting um and you feed in some kind of input into that chip which programs the cells that it's you know that are being controlled by that to give a different output than the epileptic output that we're already seeing right and so rather than just saying oh that's an epileptic fit we can predict that it's like okay yeah we can predict it but now we can actually do something to change it right we can actually change it with some kind of input I like to imagine in a very futuristic kind of application of this hopefully not too futuristic but in a very futuristic application of this you know in The Matrix when Neo is plugged in and he suddenly goes you know he his eyes go blank and they're uploading all this stuff to him and he wakes up and he goes I know Kung Fu that's sort of the the kind of very futuristic U sci-fi application of of what we're actually doing we're actually encoding Kung Fu into these cells and then trying to replicate that yeah that is absolutely fascinating yeah I was actually going to ask you sort of what are you most excited about in this field like what do you see coming ABS future for you would it be sort of the brain computer interface space I think the brain computer interface space is fascinating you know we're seeing really interesting companies even local Australian companies like synchron overtaking these multi-billion dollar uh organizations in in this uh space um we're going to see a lot more interesting kind of uh predictions of what people who might not be able to communicate normally um and and reconstructing their thoughts and and what they're actually trying to you know represent um and so that's going to transform a lot of people's lives it's going to transform medicine in that sense but what I'm much more interested in you know it's still very interesting and it's going to be really cool in the near future but what I'm much more interested in in is this idea of can we write things into neurons can we program neurons to give us the kind of outputs that we want rather than just you know interpreting what they're already doing and that's what we're doing a cical labs in the long run um you know it might take a long time to get there but really really interested to do that very cool and then something which obviously people are talking about a lot is like the ethics of AI and the explainability

and the transparency of AI which is a current problem with the you know blackbox methods that we have one of the things I found interesting about active inference is people talk about how you can actually have much more transparent and explainable AI potentially and I see what you're doing with the do you think with the Nano wi networks you also have the opportunity for that because you can actually go in and don't know we didn't get on to how you actually visualize and analyze them which is interesting but like do you think that this type of AI would be more explainable than sort of traditional artificial neural networks oh without a question I mean you know the reservoir Computing approach that I that I touched on does use that black box approach which is why I didn't really like it I used it for the first couple of um papers that I worked on and then was sort of like this is not how the brain works maybe it is maybe there are some aspects of it but the brain is not a black box I mean to us it is but you know there are so many little pieces that work together to you know create these kind of uh emerging properties and there's rules that govern that and there's structures that govern that and so there must be a way to implement those kind of rules into some into these systems and then be able to actually see them learning and that's you know that video that I showed you can see how it's changing over time which is very very different to these blackbox approaches where we don't even know why weight you know why different weight weight changing results in this kind of learning um I think in neurons you can do all sorts of cool calcium Imaging you can do infrared kind of microscopy I I'm not a biologist so I'm sorry to my team if I was wrong about that but you know we have them on these transparent cells arrays so we can actually see we can take images you know we can put them in the incubator play a bunch of long games take them out take some images see how they've shifted see what's happened and try and correlate that with the activity that's a much more transparent way of learning in my opinion very cool so it's almost like we have more energy efficiency we have more explainability and we have like these opportunities for personalization almost like with your own uh cells in the in the chip which I think it's really cool it's a it's a very exciting Direction I think for for lots of different Technologies I guess not just brain computer interfaces as well for sure and you know I think what's really interesting is how we can combine these Technologies like neuromorphics with neurons neuromorphics are trying to replicate neurons is there a way we can use neuromorphics to kind of act as like you know eyes you know our eyes do this pre-processing kind of activity where you take in an a light signal an electrical signal convert it through some kind of cell you know activity into a more understandable signal for your occipital L and your other regions of your brain can we use

neuromorphic systems like Nano networks to take in a raw input and output a more biological like signal which then neurons would be a lot more receptive to you know I said we'd really like to encode these write things into neurons and get them to do things that way but the question of how to do that is a really difficult question um otherwise people would have figured it out by now um we have a lot of theories but one of those might be as simple as let's use these technologies that are taking in simple in or complex inputs and making them into a more biological like output based on just their inherent properties and then feed that you know interface with a biological system who knows maybe that'll be a really cool way of getting that kind of encoding and writing in method so that's something that I'm really interested in in marrying those two worlds together as well okay I'm conscious I don't want to go too much over time but I do have a little bit more got a little bit more my I've actually got as much time as you need so that's very cool um anyone else please jump in with questions but did just want to touch on um something which I'd heard about with your work before where you mentioned about the quantum um aspects of the silver Nano wire Network yes not a quantum computer scientists by any means but I'm interested in this because there's sort of this debate in Neuroscience about whether or not there's a Quantum brain like the brain Quantum um maybe you know for our faction um there's been some suggestions some theories about that's how we we smell with with Quantum tunneling and the AL faction receptors but the rest of the brain people seem to be the jury seems to be still out and I'm just interested in your network seemed to use you you mentioned that there was like some Quantum tunneling forming the bridges potenti is that is that your work or was it just you know something which um is interesting but wasn't really you examined so it was actually more of my colleagues Joel and we have a paper in nature com's uh nature Communications about that so I can I can definitely share that with people if you Google my name it's the first thing that will come up um but yeah I love that we're covering all the buzz words here it's fantastic so yes the quantum tunneling or we call it electrode tunneling um we see this Behavior it's sort of the only way we can explain this um and so that's sort of why we've hypothesized and Joel's shown some evidence for this but the idea is that you start with a no Bridge here on the left for example and over time you get this fully connected bridge but that doesn't it's not like an immediate kind of response that takes time to sort of the ions flow slowly through that it's quite quick but it it's not an immediate response but what we start to see is if for example the bridge is kind of semiformed and let's imagine these two dots here aren't there these two little um balls at the end or ions aren't there you can see you start seeing these weird fluctuations of conductance before that switch is fully

formed and that's explained by this Quantum tunneling or Electro tunneling of electron jumping through the gap between the whatever leftover Gap is is there just by chance um and and and that actually accounts for really interesting uh behaviors that I haven't covered today I don't really fully understand them myself that's not my area of expertise but Joel has explored them and has some really interesting theories about how that might affect the networks one example is this idea of criticality so criticality is sort of uh the theory that you know biological systems or a lot of systems actually real world systems um have maximal information processing at the edge of criticality or the edge of chaos so uh chaotic system is one that is sort of completely unpredictable has full chaotic activity and and ordered on the other hand an ordered system is something that nothing interesting can happen because you can you know exactly what's going to happen it's full predictable you know every single aspect of it but somewhere in the the middle somewhere in between where you're not too chaotic there's still quite a lot of unpredictable Behavior but there is a predictability to that chaoticness it's called the edge of chaos that's where things seem to thrive life seems to thrive and we've seen criticality in Nano networks we've seen criticality in our neurons as well in cortical labs and one of these ideas is that potentially uh the electron tunneling might um might help tune the system into this critical State wow okay we don't really fully understand it if you're really interested in that the paper is a really cool um uh summary of that I guess I'll get the link off you and I'll make sure you put it into sounds great YouTube description um yeah because I'm really interested in sort of what these synthetic systems can then tell us in turn about the brain for those sort of things we've been uncertain about for a long time like whether or not the brain uses Quantum and it's interesting actually as well because there was a Loi chip and they used that the neuromorphic nose um and that performed quite well so it would be interesting if you could sort of combine with these type of networks that maybe do have the quantum effects and sort of test out these more quantum theories of our faction that would be pretty cool for sure I I I think there's so much you know to explore here um look I I have maybe been unreasonable to some of the neuromorphic chips like ly he in this talk I know I only briefly spoke about it there's definitely space for that they're doing amazing things um there's some really cool learning algorithms and some really awesome people working on this stuff uh I think my perspective from this is that we want to try and replicate more brain like learning and more brain-like activity and for that we need a structure that is more brain-like as well um the structure function behavior and and relationship is key in my opinion to unlock in the emergence the emergent properties in you know the brain um and so a lot of those chips lack that um and hopefully people you

know will work more and more on these brain inspired systems um and and start to consider the structural element of it as well because it is extremely important and I guess you've spoken about the nanowire networks and you've also spoken about the more neuronal networks like where's your work going the direction like are you moving away from the Nano y networks or can you see like a hybrid of of both actually being quite useful in the future yeah so I think I touched on that briefly you know marrying the two nanoes and and um and neurons together as sort of this like encoding Bridge or something like that that's you know unofficially something that I'm interested in um I'd love to explore that whether or not that's something that I'll have the time and the resources to do we'll see um but that's you know definitely where my interests lie um I'm very happy to be able to apply you know my knowledge and my expertise in these kind of algorithm development and brain learning in actual neurons so that's sort of what my focus is on at the moment but I'm still very much in touch with my supervisor and my team at University of Sydney and you know we've got a paper coming out um later this year or I guess next year um or did it just come out now I think it just recently came out actually what am I talking about gosh what what year is it it just recently came out in in in nature as well um so nature comes so yeah go check that one out as well very cool okay has anybody else got any questions before we wrap up cool okay I've got one last question one thing I'm always curious about is um if there's any sort of major problems in your field that you want someone to go out and solve almost like CU anyone watch and they might be listen they might be working on some obscure little problem and um do you have any asks of it could be you know this mathematical theorem or you know create some Hardware that we can do this weird thing with like what would be your main asks I think for anybody listening who might work on something weird and wonderful so I think for that let me just go back to the final slide again I know I'm jumping around back and forth but I think what I'm working on most at the moment and what I think is the most important thing for our work and for anyone who's working in this space is figuring out that encoding Paradigm what is it that the eyes do that the ears do that the nose does actually I think the nose does it least but that our sensory organs do so well to encode you know what might be seen as nonsense information you know uh there's obviously structures to it and and it's not nonsense but it it it's you know light or sound or very different forms of information into something that the brain can understand and make inferences about you know and not just make inferences build models of the world to try and replicate it you know active inference is all about trying to understand that um how do we do that obviously that is a huge question but in a simpler way how do we encode a simple

electrical impulse or a string a temporal sequence into something that neurons can understand okay and we can we can actually elicit different outputs from it very cool okay so if anybody's listening and they're working on please get in touch Alon because uh please do yeah that would definitely catalyze the field I guess solving those kinds of questions and problems if we can solve that then yeah I mean we'll be having very different conversations in the next few years I think very very exciting conversation yeah well hopefully the more stream will still be going we can bring you back I would love that I would absolutely love that okay right I guess we'll end there have you got any last comments or anything before we end the recording just feel free to email me reach out I think science needs to really be open and collaborative and you know I do work for a company but I'm also still very much about open source work and you know I will you and add me on LinkedIn and let's just keep chatting let's just keep answering these questions right thank you so much and thank you everybody for listening and for joining the call today and thank you for having me I'll see you later